

Multi-Agent NFL Analytics System

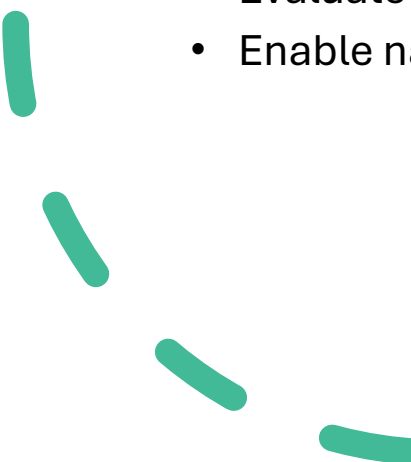
*Empowering Tactical Decision-Making
with AI Agents*

Author: Victor Santana Puga





Overview & Objectives

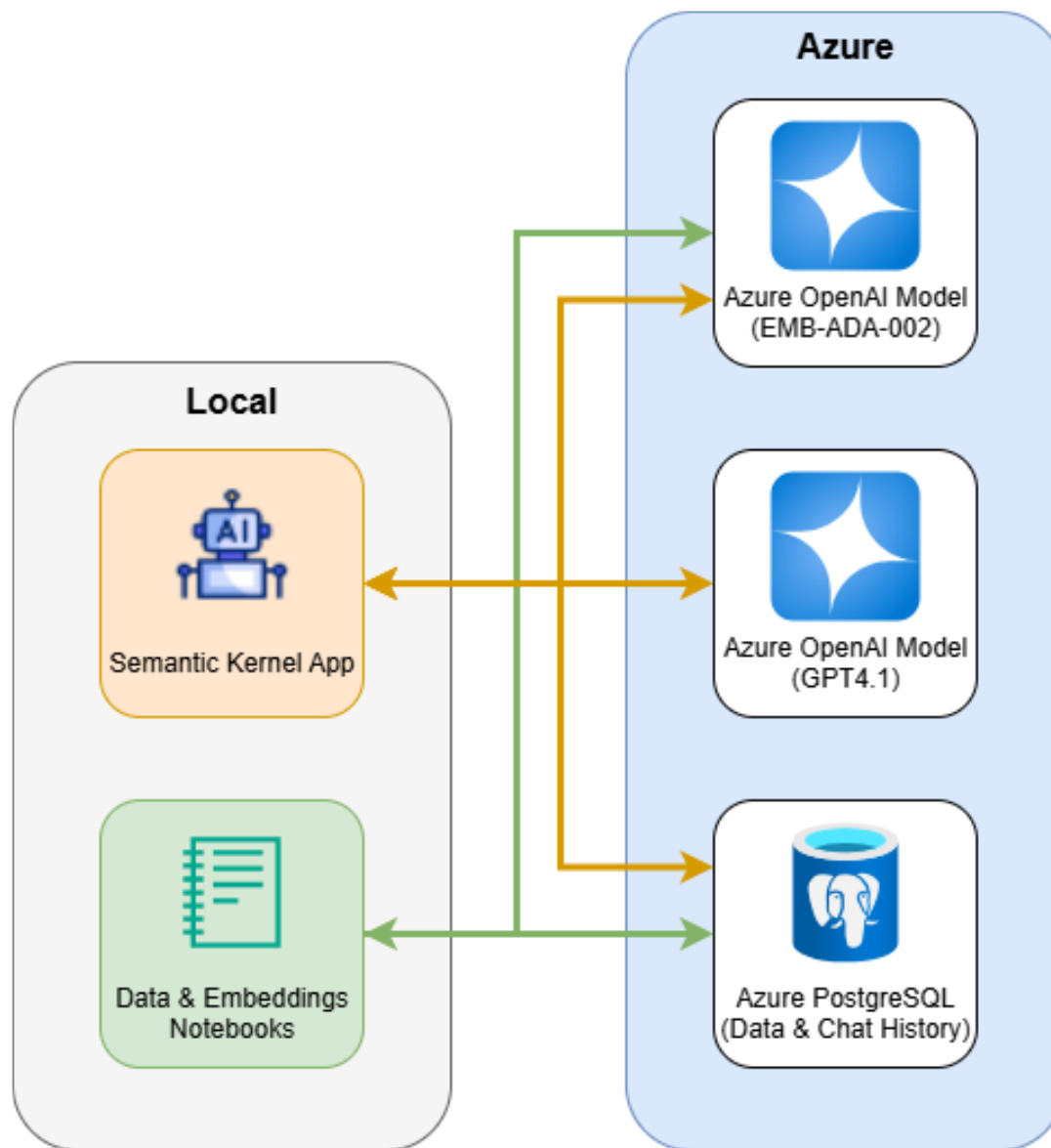
- **Overview:** This project showcases a multi-agent AI system designed to analyze NFL games, players, plays, and formations. It leverages intelligent agents to automate tactical analysis, assess strategy effectiveness, and answer complex football-related questions—empowering coaches, analysts, and decision-makers.
 - **Objectives:**
 - Facilitate faster, data-driven tactical decisions in football.
 - Automate the analysis of formations and plays using real match data.
 - Evaluate team and player performance with key metrics.
 - Enable natural language interaction with specialized agents.
- 

Use Cases

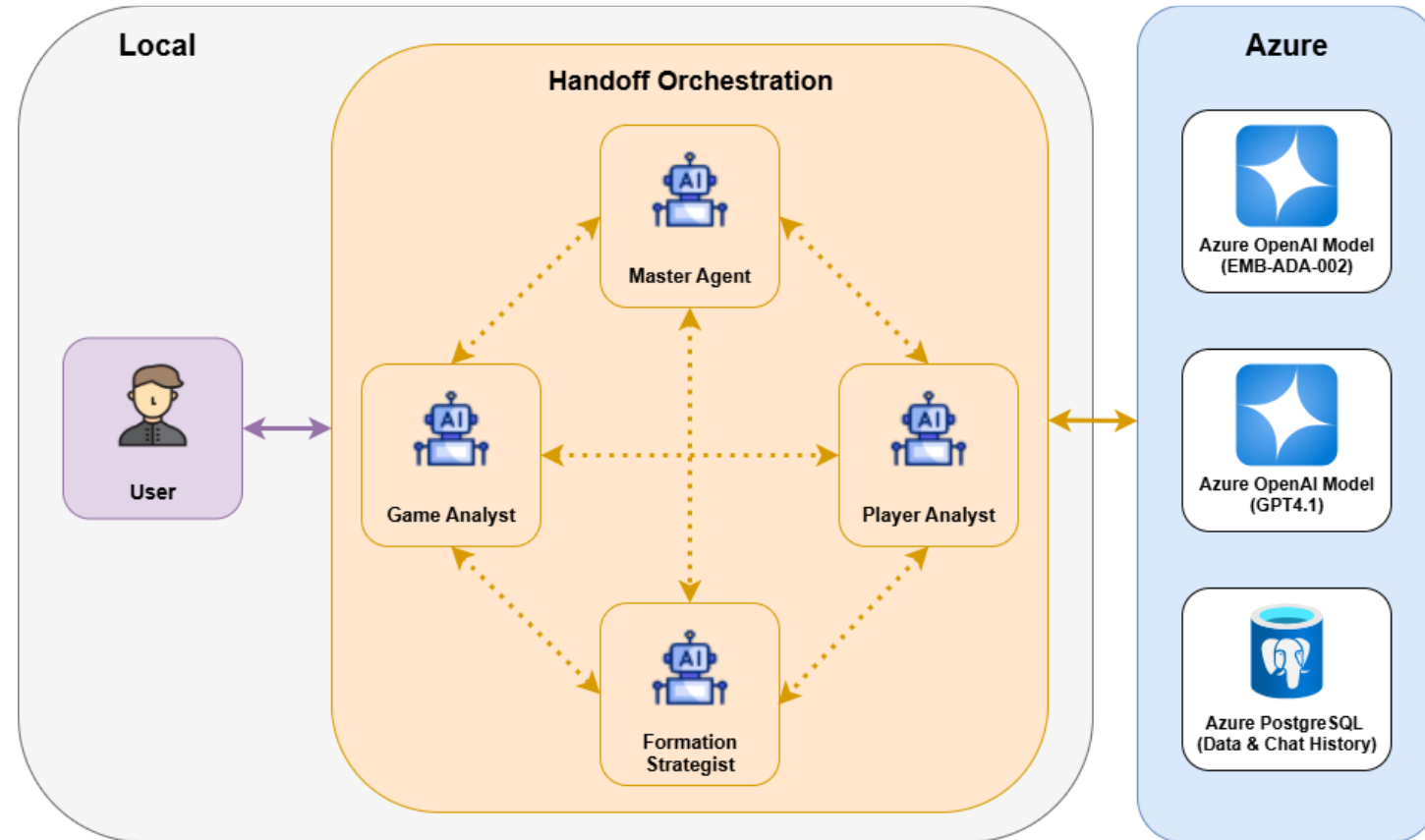
- **Analytics & Comparison:**
 - Examine player performance and game data
 - Evaluate teams and players across different games and seasons
- **Tactical Insights:**
 - Model tactical situations
 - Assess offensive and defensive setups
 - Suggest the best formations depending on the situation
- **Summarization & Interpretation:**
 - Create automated summaries of games
 - Identify and explain important moments



High Level Architecture



Multi-Agent Architecture



Agents

- **Master agent:** Orchestrates user interactions, classifies intent, and delegates tasks to specialized agents while maintaining session continuity.
- **Game Analyst:** Delivers insights from game-level data, including team performance, match summaries, and key events.
- **Player Analyst:** Analyzes individual player metrics, trends, and comparisons across games and seasons.
- **Formation Strategist:** Evaluates tactical formations and recommends optimal setups based on game context and strategy.

Data

- **Source:** Beginners Sports Analytics NFL Dataset (Kaggle)
- **Datasets:**
 - **Games:** Metadata for each NFL game (date, time, teams, week).
 - **Players:** Biographical and physical data (height, weight, position, college).
 - **Plays:** Tactical and scoring details (formations, down, yardage, EPA, penalties).
 - **Week Data:** Player tracking data per frame (position, speed, orientation, events).
- **Data Coverage:**
 - The datasets cover the **2018 NFL regular season**.
 - **Week Data** only includes tracking information for **Week 11**.



Tech Stack & Github Repo

-  **Python + Semantic Kernel Framework**

Agent definition and orchestration

-  **PostgreSQL on Azure**

Storage for datasets and chat history

-  **Azure OpenAI Models**

text-embedding-ada-002: Semantic embeddings for retrieval

GPT-4.1: LLM for agent reasoning and response generation



Github Repo: [victormspg/nfl_stats_agent](https://github.com/victormspg/nfl_stats_agent)